

COURSE INFORMATION

Description:

Learners explore Python programming by designing and implementing solutions to real-world problems. Students develop foundational programming skills using variables, data types, conditionals, loops, functions, and data structures. The course emphasizes problem-solving, debugging, and code comprehension while introducing the use of AI-assisted tools to support development. Learners practice building, evaluating, and refining programs, both independently and with AI assistance. By the end of the course, students are able to create, understand, and explain basic Python programs and apply structured thinking to solve problems.

Career Cluster: Information Technology

Instructional Level: A.A.S. - Associate in Applied Science

Total Credits: 2

Total Hours: 54

TEXTBOOKS

Learn Python Programming: A comprehensive, up-to-date, and definitive guide to learning Python - 4th ed. Edition - Fabrizio Romano/Heinrich Kruger

The Missing Tools: How Developers Learn to Think Quickly - 1st Edition - Cash Myers

Python Programming Language: a QuickStudy Laminated Reference Guide Pamphlet – Robin Nixon – (Optional)

CORE ABILITIES

1. Act Professionally

Status: Active

Criteria

- 1.1. Take responsibility for one's own actions.
- 1.2. Conform to the technical standards of a profession.
- 1.3. Practice morally responsible behavior.
- 1.4. Take responsibility for staying up-to-date
- 1.5. Maintain confidentiality.
- 1.6. Exhibit respect for people and property.
- 1.7. Exhibit a sense of global awareness.
- 1.8. Display appearance appropriate for work environment.

2. Communicate Clearly

Status: Active

Criteria

- 2.1. Speak clearly so others can understand.
- 2.2. Write with clarity so that others can understand.
- 2.3. Select appropriate means to convey a message.
- 2.4. Ask questions for clarification.
- 2.5. Participate in discussions and group presentations.
- 2.6. Interpret nonverbal communications.
- 2.7. Use active listening skills.
- 2.8. Apply standards of spelling, English, grammar, and punctuation.

3. Solve Problems

Status: Active

Criteria

- 3.1. Use critical, creative thinking skills.
- 3.2. Apply problem solving steps.
- 3.3. Demonstrate open-mindedness.
- 3.4. Evaluate alternatives to choosing a solution.
- 3.5. Implement solutions appropriately.
- 3.6. Use research to solve problems.
- 3.7. Use appropriate mathematical calculations.

4. Value Learning

Status: Active

Criteria

- 4.1. Assume responsibility for lifelong learning.
- 4.2. Identify own learning needs.
- 4.3. Apply effective learning processes.
- 4.4. Help other to learn effectively.

5. Work Cooperatively

Status: Active

Criteria

- 5.1. Demonstrate respect for self and others.
- 5.2. Contribute to a group activity with ideas, suggestions, and effort.
- 5.3. Complete own share of tasks necessary to finish a group project.
- 5.4. Maintain a safe and healthy work environment for self/group.
- 5.5. Set goals, standards, limits, for self/group.
- 5.6. Display effective interpersonal skills.
- 5.7. Resolve conflicts in a constructive manner.
- 5.8. Seek help when needed.
- 5.9. Demonstrate ability to work with a diverse population.

6. Work Productively

Status: Active

Criteria

- 6.1. Attend regularly and on time.
- 6.2. Exhibit organizational skills.
- 6.3. Locate resources for problem solving.
- 6.4. Display productive work ethic.
- 6.5. Maintain necessary knowledge and skills.
- 6.6. Use effective, efficient processes.
- 6.7. Show self-direction in starting tasks.
- 6.8. Demonstrate reliability.
- 6.9. Follow directions.
- 6.10. Complete required tasks on time.

COURSE COMPETENCIES

1. Analyze problems and define program logic

Domain: Cognitive Level: Analyzing Status: WIP

Assessment Strategies

- 1.1. Project assignments and quiz

Criteria

- 1.1. Break problems into step-by-step logic
- 1.2. Identify inputs, outputs, and constraints
- 1.3. Represent solutions using structured logic (pseudocode or code)
- 1.4. Distinguish between clear and ambiguous problem statements

Learning Objectives

- 1.a. Define program logic and problem decomposition
- 1.b. Identify inputs, outputs, and constraints
- 1.c. Describe step-by-step problem-solving approaches

2. Implement Python programs using core programming constructs

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

- 2.1. Project assignments and quiz

Criteria

- 2.1. Write Python programs using variables, data types, conditionals, and loops
- 2.2. Use lists and dictionaries to manage data
- 2.3. Define and use functions
- 2.4. Apply basic file input/output operations
- 2.5. Produce readable and structured code

Learning Objectives

- 2.a. Identify Python syntax and constructs

2.b. Explain variables, data types, and control structures

2.c. Describe functions and basic data structures

3. Utilize AI-assisted tools to develop and refine Python programs

Domain: Cognitive Level: Applying Status: WIP

Assessment Strategies

3.1. Project assignments and quiz

Criteria

3.1. Use AI tools to generate or assist in writing code

3.2. Provide clear prompts that describe the intended solution

3.3. Modify and adapt AI-generated code to meet requirements

3.4. Integrate AI-generated components into working programs

Learning Objectives

3.a. Describe how AI tools support programming

3.b. Identify effective prompting strategies

3.c. Explain how to adapt generated code

4. Evaluate and debug Python programs for correctness and clarity

Domain: Cognitive Level: Evaluating Status: WIP

Assessment Strategies

4.1. Project assignments and quiz

Criteria

4.1. Identify and fix syntax and logical errors

4.2. Test programs using sample inputs

4.3. Explain how a program works step-by-step

4.4. Compare different approaches to solving the same problem

Learning Objectives

4.a. Define debugging and testing

4.b. Identify common programming errors

4.c. Explain program execution flow

5. Adapt programming approaches with and without AI assistance

Domain: Cognitive Level: Creating Status: WIP

Assessment Strategies

5.1. Project assignments and quiz

Criteria

5.1. Write basic programs without relying on AI tools

5.2. Use AI to improve or extend existing solutions

5.3. Explain their code to others clearly

5.4. Adjust solutions based on constraints or requirements

Learning Objectives

- 5.a. Describe differences between manual and AI-assisted development
- 5.b. Explain when AI use is appropriate
- 5.c. Identify strategies for adapting solutions